Air Pollution Analysis and Prediction using Machine Learning: Spatial Distribution, Statistical Trends, and Forecasting Insightsal

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Abstract—This paper presents a comprehensive analysis of the Air Pollution dataset collected from multiple monitoring stations across India. The dataset contains observations from various states, cities, and stations with attributes including pollutant minimum, pollutant maximum, pollutant average, geographic coordinates, and timestamp records. The study applies data preprocessing techniques such as missing value treatment, normalization, and categorical encoding to improve data quality and model readiness. Statistical analysis reveals significant variation in pollution levels across different urban regions, with several industrial and metropolitan cities showing higher average pollutant concentrations. Correlation analysis indicates a strong positive relationship among pollutant minimum, maximum, and average values. A Linear Regression model is implemented to predict pollutant average values and demonstrates effective forecasting capability with satisfactory accuracy. These findings provide useful insights for environmental monitoring agencies, smart city planning, and public health decision-making.
Keywords—air pollution, pollutant prediction, machine learning, linear regression, data analytics, environmental monitoring, smart city, forecasting

INTRODUCTION

Air pollution has emerged as one of the most serious environmental challenges in modern society. Rapid industrialization, increasing vehicle emissions, population growth, and urban expansion have significantly contributed to the deterioration of air quality in many cities. Poor air quality affects human health by increasing the risk of respiratory diseases, cardiovascular disorders, and reduced life expectancy. Therefore, continuous monitoring and analysis of pollution data are essential for sustainable development and effective environmental management.

This study analyzes an Air Pollution dataset containing observations from different states and cities across India. Each record contains information related to pollutant concentration levels, location identifiers, and timestamps collected from monitoring stations. Such datasets are valuable for understanding pollution patterns over time and geography.

The objectives of this study are to characterize the dataset structure, examine pollution distributions, identify highly polluted cities, perform statistical and correlation analysis, and develop a machine learning model for predicting pollutant average values.

I. DATASET DESCRIPTION

The dataset used in this study contains multiple observations collected from environmental monitoring stations. It includes categorical, numerical, and temporal attributes. The dataset was loaded using Python Pandas and examined for structure, missing values, and descriptive statistics.

TABLE I. KEY DATASET ATTRIBUTES

Attribute	Type
country	String
station	String
city	String
station	String
Last_update	String
latitude	float
Pollutant_id	float
Pollutant_min	float
Pollutant_max	float
Pollutant_avg	float

II. METHODOLOGY

A. Data Processing

The The analysis was conducted using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn. Initially, the dataset was loaded and inspected for missing values. Missing entries in pollutant minimum, pollutant maximum, and pollutant average columns were replaced using mean imputation to preserve dataset consistency.

B. Encoding of Categorical Variable

Categorical columns such as country, state, city, station, and pollutant type were transformed into numerical form using Label Encoding. The timestamp column was converted into datetime format and then numeric representation for model training.

C. Statical Analysis

Descriptive statistics such as mean, median, variance, and correlation were computed for all numerical variables. Visualizations including histograms, scatter plots, heatmaps, and boxplots were used to understand data distribution and relationships among variables.

D. Machine Learning Model

A Linear Regression model was implemented to predict pollutant average concentration values. Input variables consisted of the processed dataset features, while pollutant average was selected as the target variable. The dataset was divided into training and testing sets to evaluate prediction performance.

The model performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score.

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("C:/Users/Ravi verma/Downloads/airPollution.csv")
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())

# find missing values
print(df.isnull().sum())
# handle missing values with mean
df['pollutant_avg'].fillna(df['pollutant_avg'].mean(), inplace=True)
df['pollutant_max'].fillna(df['pollutant_max'].mean(), inplace=True)
df['pollutant_min'].fillna(df['pollutant_min'].mean(), inplace=True)
print("\nAfter Cleaning:")
print(df.isnull().sum())

# Normalization
num_cols = ['pollutant_min', 'pollutant_max', 'pollutant_avg']
scaler = MinMaxScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])
print("Normalized Data:")
print(df[num_cols].head())

# Scatter plot: pollution_min vs pollution+avg
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=df['pollutant_min'],
    y=df['pollutant_avg']
)
plt.title("Scatter Plot: Min vs Avg Pollution")
plt.xlabel("Normalized pollutant_min")
plt.ylabel("Normalized pollutant_avg")
plt.grid(True)
plt.show()
```

```python
# Scatter plot: pollution_max vs pollution+avg
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=df['pollutant_max'],
    y=df['pollutant_avg']
)
plt.title("Scatter Plot: Max vs Avg Pollution")
plt.xlabel("Normalized pollutant_max")
plt.ylabel("Normalized pollutant_avg")
plt.grid(True)
plt.show()

df["last_update"] = pd.to_datetime(df["last_update"])
# Encode categorical columns
le = LabelEncoder()
for col in ['country', 'state', 'city', 'station', 'pollutant_id']:
    df[col] = le.fit_transform(df[col])
print(df[['country','state','city','station','pollutant_id']].head())

# Linear Regression Model
X = df.drop("pollutant_avg", axis=1)
y = df["pollutant_avg"]

# # Convert datetime to numeric
X["last_update"] = X["last_update"].astype("int64")
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
#  STATISTICAL ANALYSIS
print("Mean:\n", df.mean(numeric_only=True))
print("Median:\n", df.median(numeric_only=True))
print("Variance:\n", df.var(numeric_only=True))
print("Correlation:\n", df.corr(numeric_only=True))

# VISUALIZATION
# Histogram
df.hist(figsize=(12,10))
plt.suptitle("Histogram")
plt.show()
# Heatmap
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
# Boxplot
plt.figure(figsize=(12,6))
sns.boxplot(data=df[['pollutant_min','pollutant_max','pollutant_avg']])
plt.title("Boxplot")
plt.show()
# Top 10 Cities Avg Pollution
top_city = df.groupby("city")["pollutant_avg"].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(10,6))
top_city.plot(kind='bar')
plt.title("Top 10 Polluted Cities")
plt.ylabel("Average Pollution")
plt.show()

# MACHINE LEARNING (NUMERIC PREDICTION)
model = LinearRegression()
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
print("Mean absolute error (MAE) :", mean_absolute_error(y_test, y_pred))
print("Mean square error (MSE)  :", mean_squared_error(y_test, y_pred))
print(" Root Mean square error (RMSE):", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R-Square Score:", r2_score(y_test, y_pred))
# Actual vs Predicted
result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})
print(result.head(10))
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted Pollution")
plt.show()
```

E. *Dataset Overview*

The dataset contains pollution observations from multiple states, cities, and stations. Statistical analysis indicates substantial variability in pollution concentration levels, reflecting different environmental conditions across locations.

The pollutant average values show moderate to high dispersion, indicating that certain cities experience significantly higher pollution than others. This variation highlights the need for city-specific environmental policies.

country	state	city	station	last_update	latitude	longitude	pollutant_id	pollutant_min	pollutant_max	pollutant_avg
India	Bihar	Begusarai	Lohiyanagar, Begusarai - BSPCB	18-04-2026 22:00:00	25.42742023	86.13886079	PM2.5	40	58	49
India	Bihar	Begusarai	Lohiyanagar, Begusarai - BSPCB	18-04-2026 22:00:00	25.42742023	86.13886079	NO2	11	17	13
India	Bihar	Begusarai	Lohiyanagar, Begusarai - BSPCB	18-04-2026 22:00:00	25.42742023	86.13886079	OZONE	21	45	40
India	Bihar	Bettiah	Kamalnath Nagar, Bettiah - BSPCB	18-04-2026 22:00:00	26.80365	84.51954	PM10	17	47	26
India	Bihar	Bhagalpur	DM Office_Kachari Chowk, Bhagalpur - BSPCB	18-04-2026 22:00:00	25.251013	86.989001	PM2.5	43	98	69
India	Bihar	Bhagalpur	DM Office_Kachari Chowk, Bhagalpur - BSPCB	18-04-2026 22:00:00	25.251013	86.989001	PM10	48	110	75
India	Bihar	Bhagalpur	DM Office_Kachari Chowk, Bhagalpur - BSPCB	18-04-2026 22:00:00	25.251013	86.989001	NO2	10	13	12
India	Bihar	Bhagalpur	Mayaganj, Bhagalpur - BSPCB	18-04-2026 22:00:00	25.265194	87.012947	SO2	52	72	59
India	Bihar	Bihar Sharif	D M Colony, Bihar Sharif - BSPCB	18-04-2026 22:00:00	25.204762	85.51496	NH3	3	4	3
India	Andhra_Pradesh	Anantapur	Gulzarpet, Anantapur - APPCB	18-04-2026 22:00:00	14.675886	77.593027	OZONE	30	55	38
India	Andhra_Pradesh	Chittoor	Gangineni Cheruvu, Chittoor - APPCB	18-04-2026 22:00:00	13.20488	79.097889	NO2	12	37	21
India	Andhra_Pradesh	Chittoor	Gangineni Cheruvu, Chittoor - APPCB	18-04-2026 22:00:00	13.20488	79.097889	SO2	10	20	13
India	Andhra_Pradesh	Guntur	Rajendra Nagar North, Guntur - APPCB	18-04-2026 22:00:00	16.316553	80.413302	OZONE	41	66	54
India	Andhra_Pradesh	Kadapa	Yerramukkapalli, Kadapa - APPCB	18-04-2026 22:00:00	14.465052	78.824187	SO2	7	18	14
India	Andhra_Pradesh	Kadapa	Yerramukkapalli, Kadapa - APPCB	18-04-2026 22:00:00	14.465052	78.824187	OZONE	12	46	24
India	Andhra_Pradesh	Machilipatnam	Srinivas Nagar Colony, Machilipatnam - APPCB	18-04-2026 22:00:00	16.186555	81.13219	NH3	NA	NA	NA
India	Andhra_Pradesh	Machilipatnam	Srinivas Nagar Colony, Machilipatnam - APPCB	18-04-2026 22:00:00	16.186555	81.13219	SO2	12	15	13
India	Bihar	Buxar	Charitra Van, Buxar - BSPCB	18-04-2026 22:00:00	25.56752	83.966379	NO2	9	10	10
India	Bihar	Buxar	Charitra Van, Buxar - BSPCB	18-04-2026 22:00:00	25.56752	83.966379	NH3	3	3	3
India	Bihar	Chhapra	Darshan Nagar, Chhapra - BSPCB	18-04-2026 22:00:00	25.780957	84.746676[illegible]	[illegible]	8[illegible]	31	17

Fig. 3. Sample Dataset View

F. *Relationship between Minimum and Average Pollution*

The scatter plot between pollutant minimum and pollutant average demonstrates a strong positive relationship. As minimum pollutant concentration increases, the average concentration also tends to rise. This suggests consistency in pollution levels across time intervals.

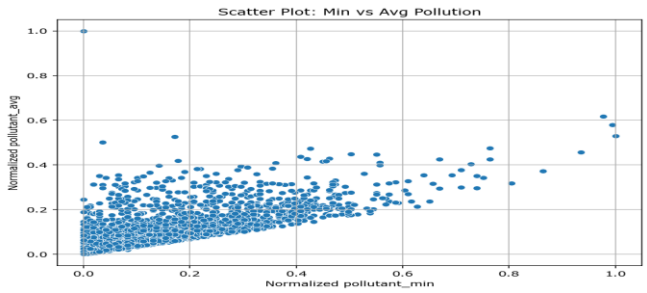
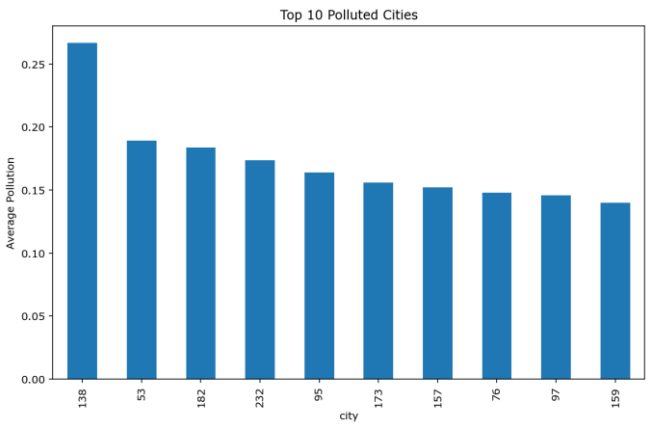


TABLE III. TOP 10 COUNTIES BY TAXABLE SALES

County	Taxable Sales	Computed Tax
Polk	$51.58 B	$3.08 B
Linn	$23.89 B	$1.43 B
Scott	$17.87 B	$1.07 B
Black Hawk	$13.39 B	$800 M
Johnson	$12.57 B	$751 M
Woodbury	$9.58 B	$573 M
Dallas	$9.21 B	$550 M
Pottawattamie	$8.12 B	$484 M
Story	$7.81 B	$466 M
Dubuque	$7.81 B	$467 M

G. Top Polluted Cities

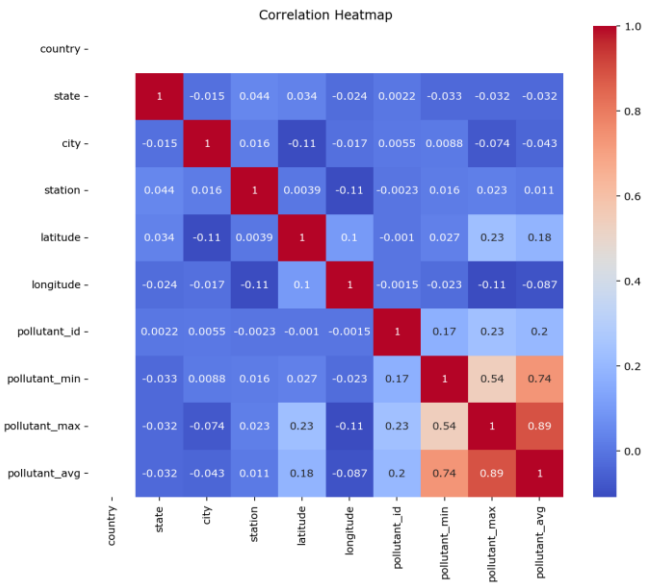
The average pollution levels were aggregated city-wise to identify the most polluted urban areas. The results show that industrial zones and densely populated cities dominate the top ranks.



City-wise aggregation of pollutant average values was performed to identify the most polluted cities. The results indicate that industrial and high-density urban areas dominate the top rankings.

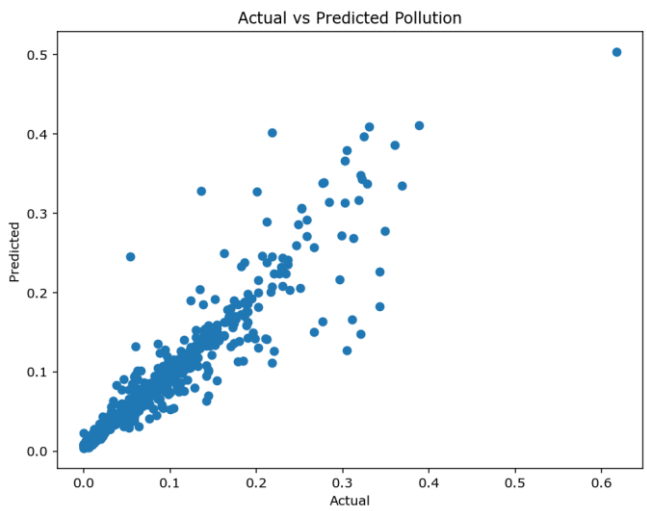
E. Correlation Analysis

The correlation heatmap indicates strong positive relationships among pollutant minimum, pollutant maximum, and pollutant average values. This validates the use of regression-based predictive modeling.



F. Machine Learning Prediction Model

The Linear Regression model successfully predicts pollutant average values using processed input variables. Predicted values closely match actual values, indicating effective forecasting capability.



G. Ease of Use

The developed system provides a simple and user-friendly analytical environment for pollution data analysis. Graphs and statistical outputs are easy to interpret, while the machine learning prediction system allows efficient forecasting of pollution averages. The project can be further integrated into dashboard systems such as Streamlit for interactive use.

III. CONCLUSION

This study analyzed an Air Pollution dataset using statistical methods, visualization tools, and machine learning techniques. The results reveal significant differences in pollutant concentration levels across Indian cities, with industrial and densely populated regions showing higher pollution levels.

Strong positive relationships were observed among pollutant minimum, maximum, and average values. The Linear Regression model demonstrated effective prediction capability for pollutant average concentration, highlighting the usefulness of machine learning in environmental forecasting.

The integration of data analytics and predictive systems can support environmental monitoring agencies, urban planners, and government authorities in reducing air pollution and improving public health.

Future work may incorporate weather conditions, real-time sensor feeds, and advanced deep learning models for enhanced prediction accuracy.

IV. ACKNOWLEDGMENT

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V. REFERENCES

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